

# Spatiotemporal data modeling and prediction algorithms in intelligent management systems

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## ABSTRACT

In order to solve the problem of difficulty in learning semantic pattern representations between user dynamic interest sequences using path based and knowledge graph based entity embedding methods, the author proposes research on spatiotemporal data modeling and prediction algorithms in intelligent management systems. The author first makes a preliminary analysis of the wireless network data (mainly the data of cellular mobile networks) obtained by Internet service providers, reveals that the data of adjacent base stations have temporal and spatial correlations, then establishes a hybrid deep learning model for spatio-temporal prediction, and proposes a new spatial model training algorithm. Finally, experiments were conducted using wireless network datasets to evaluate the performance of the model. The experimental results show that based on data analysis, it can be seen that the prediction of the system has effectively improved by 99 %.

**Conclusion:** The spatiotemporal data modeling and prediction algorithm proposed by the author in the intelligent management system significantly improves prediction accuracy.

## 1. Introduction

With the continuous popularization and development of intelligent devices and technology, more and more spatiotemporal data has become easier to obtain, and it plays an unprecedented application value in different fields, such as environmental protection, transportation, smart healthcare, epidemic prevention and control, etc [1]. Spatiotemporal data is a structured data that contains spatiotemporal attributes, and spatiotemporal is the abbreviation of time and space [2,3]. Time and space refer to a sequence of things, with time commonly used to describe the sequence of events and space commonly used to describe the position of objects. Unlike conventional structured data that is often based on the fundamental assumption that data instances have independent and identically distributed relationships, spatiotemporal data has always attracted researchers in the fields of data mining and machine learning to continuously explore due to its complex and difficult to directly represent relationships between spatiotemporal dimensions [4].

Through intelligent management systems, comprehensive automation management can be achieved, making management behavior more precise, efficient, and intelligent [5]. In intelligent management systems, personalized management has become an important topic.

Through artificial intelligence technology, intelligent management systems can simulate human thinking and behavior, thereby achieving more intelligent management [6]. Personalized intelligent management can not only improve management efficiency and efficiency, but also better meet the needs and requirements of different users. Intelligent management systems have become an important means of modern management, and their application scope is becoming increasingly widespread, involving various industries and fields. Among them, big data and the Internet of Things are important components of intelligent management systems [7].

## 2. Literature review

Predictive modeling of spatiotemporal data is a fundamental research topic in spatiotemporal data mining. Its goal is to predict future observations of spatiotemporal data based on historical data. The types of specific spatiotemporal data prediction modeling problems often vary depending on different application scenarios [8]. Point data, also known as event data, is typically isolated point data that is aggregated according to temporal or spatial relationships to form a point sequence or event sequence [9]. Point sequence data is a widely used data type in

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spatiotemporal data prediction modeling, often used to predict the crime situation in a region. Bharathi, D. et al. used models based on traffic flow theory to consider the spatiotemporal evolution of public transportation travel time. The literature mainly studied first-order models (also known as first-order models), while second-order models (usually referred to as high-order models) have not been fully explored due to their complex structure, need for calibration of parameters, difficulties in data acquisition, and difficulties in discretization and solution [10]. Subramanian, S. et al. proposed a three-dimensional spatiotemporal prediction model that can provide reliable predictions of various atmospheric parameters. This model uses discrete atmospheric data that varies with latitude, longitude, altitude, and time, and predicts altitude profiles of atmospheric parameters for future time instances [11]. Fang, Q. et al. investigated the spatiotemporal patterns of urban bus traveler activity using a one month bus trajectory dataset based on smart card data in Kumamoto City. Based on the concept of compact cities, the ordinary least squares regression model was used to explore the factors that affect the passenger volume of public transportation at the level of 17 important core areas of Kumamoto City's bus stops. Then, using the Geographically Weighted Regression (GWR) model, the spatial heterogeneity of bus passenger volume can be obtained, and the spatial distribution of parameter estimation can be visualized. By comparing the results of the two models, it was found that they performed similarly in terms of global fitting and interpretation accuracy [12].

By utilizing large-scale system data, the author proposes a deep learning method for wireless network spatiotemporal modeling and prediction, and conducts performance evaluation through experiments.

### 3. Method

#### 3.1. Scale characteristics of spatiotemporal sequence data

Scale is a fundamental tool in geospatial analysis and a major attribute of spatial data. It is one of the main factors in spatial analysis and management [13,14]. There are differences in terms and connotations of scale in different fields. For example, in cartography, scale is defined using a mapping scale, which refers to the ratio of the distance on the map to the actual distance; In remote sensing, scale is defined using resolution, which describes the level of measurement precision. The scale studied by the author refers to the usual spatial and temporal scales, which refer to the spatial or temporal units used in the study of a certain object or process, and can also refer to the scope of a certain object or process in space or time. In spatiotemporal data mining, spatial units and time units are commonly expressed in terms of spatial and temporal granularity, while spatial and temporal ranges are described in terms of spatial breadth and temporal breadth. Geographical objects or phenomena exhibit different forms and reflect different patterns at different spatial breadth and granularity. Large scale data occupies a relatively large spatial range relative to the Earth's surface or other secondary regions that are commonly referenced in space, reflecting the overall, abstract, and contour trends of geographic objects or phenomena; On the other hand, small-scale occupies a smaller spatial range, reflecting the detailed and specific content of geological processes. Therefore, increasing breadth, reducing granularity, and roughly summarizing spatiotemporal data reflect more significant global features; On the contrary, reducing breadth and increasing granularity will result in detailed and specific spatiotemporal data, reflecting more significant local information [15].

#### 3.2. Spatial scale transformation of spatiotemporal sequence data

Space scale conversion is the process of converting data or information from one scale to another, which can be divided into scale up and scale down based on the direction of conversion. Scale up is the process of inferring larger scales from smaller scales, while scale down is the process of inferring smaller scales from larger scales [16]. The classic

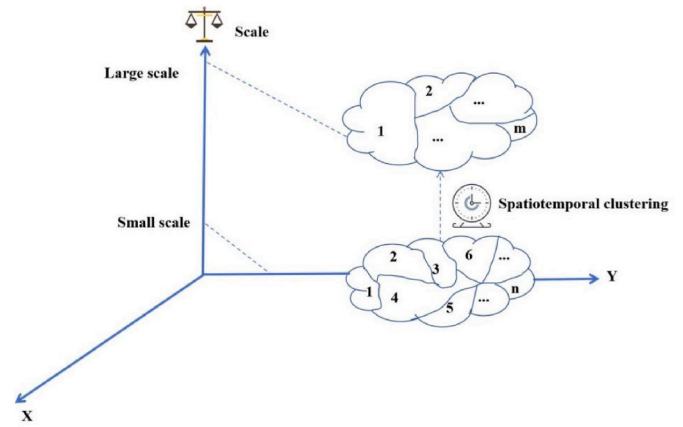


Fig. 1. Clustering based Scale Conversion Process.

methods for converting spatial scales mainly include stacking points and polygons, region weighting, maximizing preservation, and modified region weighting. In spatiotemporal data mining, clustering analysis techniques can merge small grained spatial units or objects before clustering into larger grained spatial units after clustering, and use statistical values to describe the characteristics of clusters. Therefore, clustering methods can be used for spatial scale transformation. Spatiotemporal sequence clustering analysis divides spatiotemporal sequence data into different clusters based on the adjacency of spatial units in the spatial domain and the similarity in the temporal domain, so that, under the condition of satisfying spatial adjacency, the similarity within clusters is as large as possible and the similarity between clusters is as small as possible [17]. Spatiotemporal sequence clustering analysis is different from traditional spatial point clustering. Traditional spatial point entity clustering only considers the spatial attributes between objects, while spatiotemporal sequence clustering requires measuring the similarity of time series between objects while satisfying spatial adjacency conditions. The scale up process based on cluster analysis is shown in Fig. 1. It is possible to set a small-scale spatiotemporal sequence corresponding to  $n$  spatial units and  $n$  time series, and perform spatiotemporal sequence clustering analysis on them to obtain  $m$  ( $m \leq n$ ) clusters. Several spatial units belonging to the same cluster and their corresponding time series are merged into one object and one corresponding sequence at a large scale, resulting in a spatiotemporal sequence at a large scale ( $m$  spatial units and  $m$  time series formed after clustering). This completes the process of extrapolating the spatiotemporal sequence from a small scale to a large scale. Small scale refers to the observed data scale of spatiotemporal data, while large scale refers to the data scale obtained by using clustering effectiveness evaluation indicators to select the best clustering results and scale up [18].

The modeling process is shown in Fig. 2. The original spatiotemporal sequence data is generally regarded as data  $S^s$  at a small scale, and the spatiotemporal sequence  $S^L$  at a large scale is obtained by scaling up. Modeling the sequence  $S^L$  at a large scale yields a fitted sequence  $\hat{S}^L$ , where  $\hat{S}^L$  reflects the trend of each corresponding sequence  $S^s$  at a small scale; Modeling the deviation part of  $\hat{S}^L$  relative to  $S^s$ , namely  $\varepsilon^s$ , and correcting  $\hat{S}^L$ , can obtain the prediction models  $\hat{S}^s$  for each series at a small scale. Obtain the predicted values of each spatial unit at a small scale through a small-scale prediction model. By weighted averaging the predicted values of each unit within the same cluster at a small scale, the predicted values of the corresponding spatial units at a large scale can be obtained [19].

#### 3.3. Dataset analysis

Firstly, introduce the dataset used for analysis and evaluation, and

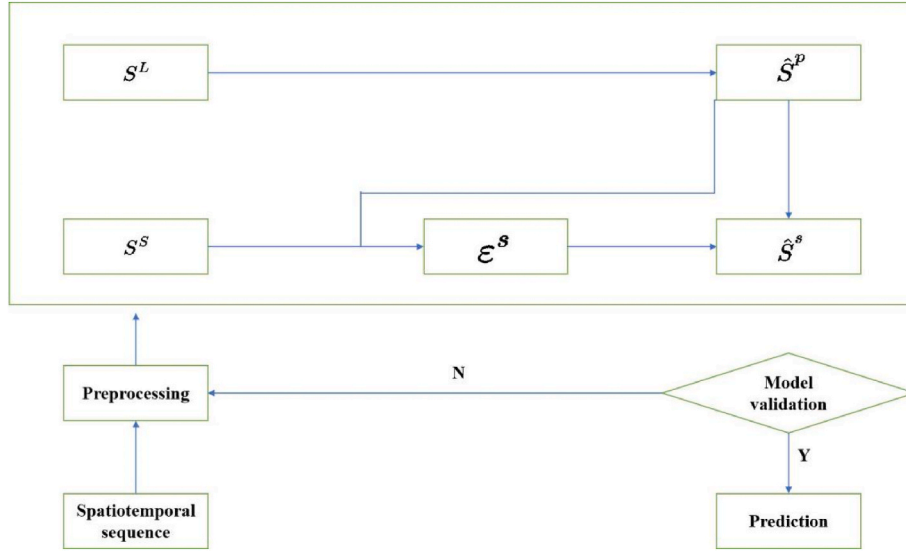


Fig. 2. General framework for predicting modeling of fused spatial scale characteristics.

then conduct a preliminary analysis of the data. The dataset used by the author is a large wireless network from a telecommunications company, which includes data generated by 2844 base stations. This dataset contains the average hourly traffic load from August 1st to December 30th. In order to simplify the analysis process, the target area is divided into grids, each containing a base station, and the grid coordinates are represented by a binary  $(m, n)$ . Use  $D = \{d_{m,n,r}\}$  to represent the data flow of grid  $(m, n)$  at time  $t$  [20]. The goal of conducting a preliminary analysis of the dataset is to explore the temporal and spatial correlations of the data. The temporal autocorrelation and spatial correlation of data can be tested separately in the temporal and spatial domains. According to the analysis, dataset D shows non-zero autocorrelation in the time domain. Use the sample autocorrelation function [1] to detect the correlation of data in the time domain, which is defined as follows:

$$r_k = \frac{\frac{1}{T} \sum_{t=1}^{T-k} (y_t - \bar{y})(y_{t+k} - \bar{y})}{c_0} \quad (1)$$

This function detects the correlation between data samples  $y_t$  and  $y_{t+k}$ , where  $\bar{y}$  is the mean and  $c_0$  is the variance of the data samples. If  $r_k$  equals  $1(-1)$ , then the data has positive (negative) autocorrelation; If  $r_k$  is equal to zero, the data does not have autocorrelation. After substituting the data samples from the dataset into formula (1), the result shows that dataset D exhibits non-zero autocorrelation in the time domain. Next, formula (2) is used to detect the correlation of the dataset in the spatial domain.

$$r = \frac{\sum_{i=0}^N (d_i - \bar{d})(d_i - \bar{d})}{(N-1) \sum_{i=0}^N \sqrt{\frac{(d_i - \bar{d})^2}{N-1}} \sqrt{\frac{(d_i - \bar{d})^2}{N-1}}} \quad (2)$$

Among them,  $N$  is the number of grids. By studying the spatial correlation between grid data, it was found through calculation that more than 50 % of the correlation values are greater than 0.4. In addition, we can see that the correlation values between different grids vary greatly, indicating that the data has spatial correlation.

### 3.4. Spatiotemporal modeling and prediction based on deep learning

From the analysis of the dataset in the previous section, it can be seen that the data has strong spatial correlation, so time modeling cannot be done solely on historical data. Based on the above observations, a new hybrid deep learning model is designed to perform spatiotemporal modeling and prediction on each grid  $(m, n)$  by utilizing historical data

collected from the target grid and its neighboring grids. This model consists of three main components: Local stack autoencoder (LSA), global stack autoencoder (GSA), and long short-term memory network (LSMN). The workflow of this model is as follows.

- 1) The data from the grid and its neighboring grids form a data patch, and GSA takes the data patch as input to generate a global encoded representation.
- 2) After GSA encoding, each data patch is transferred to the corresponding LSA and a local encoding representation is generated. Splicing the global encoding representation and local encoding representation to generate a unique encoding representation of the data patch.
- 3) Send the unique encoded representation of the data patch to LSMN for prediction. For spatial modeling, autoencoder (AE) is the best choice for data compression. AE is a neural network that typically has a hidden layer and obtains a representation of input data through training, a process known as encoding. The process of reconstructing input is called decoding. By reconstructing partially damaged inputs, denoising autoencoders make the representation of data more robust. Autoencoders can be stacked to form a deep learning network. Stacked self coding can effectively generate non-linear representations of data. Due to the positional and spatial correlations of data slices, it is necessary to obtain better encoding representations with smaller reconstruction losses. In order to achieve this goal, LSA and trained GSA are used to capture the local position dependency and spatial correlation of data slices to generate better encoding representations. Given the trained GSA, using LSA can further reduce the reconstruction loss of data slices. Firstly, train the weights of the LSA layer 1 network to reduce the reconstruction loss of the first layer in the GSA network. Then, train the higher layers of LSA to learn different representations of the lower layers. Finally, concatenate the highest level encoding representation of LSA with the highest level encoding representation of GSA to generate a unique encoding representation of the data slice. The efficient SdA-3 algorithm can be used for training GSA neural networks, but it cannot be used to train LSA. Next, define the encoding function  $enf_{ij}(\bullet)$  and decoding function  $def_{ij}(\bullet)$  for the  $i$ -th layer in the LSA network, as shown in formulas (3) and (4), respectively.

$$enf_{ij}(X_{ij}) = Y_{ij} = \delta(W_{ij}X_{ij} + b_{ij}) \quad (3)$$

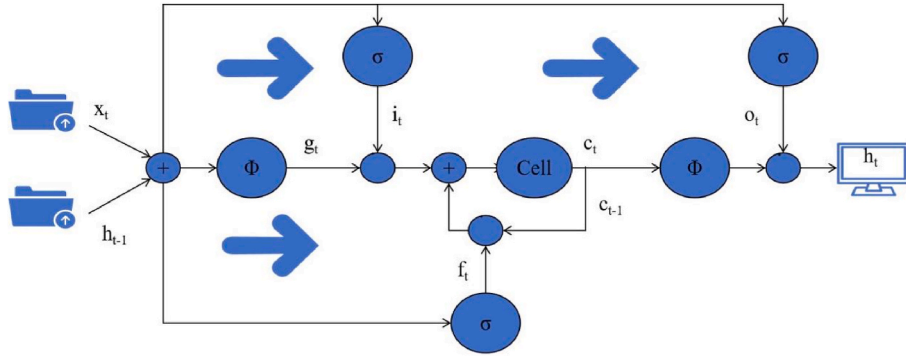


Fig. 3. Lsmn unit structure.

$$del_{l_j}(Y_{l_j}) = \begin{cases} \delta(W_{g1}^* (W_{g1}^* X_{l_1} + b_{g1}^*) + b_{g1}^* + W_{l_1}' Y_{l_1} + b_{l_1}'), & j = 1 \\ \delta(W_{l_j}' Y_{l_j} + b_{l_j}'), & \text{otherwise} \end{cases} \quad (4)$$

$\delta(\cdot)$  is the activation function, and the author uses the sigmoid function as the activation function;  $X_{l_j}$  is the input of layer  $j$ ;  $Y_{l_j}$  is the encoding result of the  $j$ th layer, and  $Y_{l_j}$  can also serve as the input of the upper layer, that is  $Y_{l_j} = X_{l_{j+1}}$ .  $W_{l_j}$  and  $W_{l_j}'$  are the weights for encoding and decoding, respectively, while  $b_{l_j}$  and  $b_{l_j}'$  are the bias for encoding and decoding.

When training GSA and LSA, use tiedweight, which means that the weight matrix in the decoding function is a transpose of the weight matrix in the encoding function.  $W_{l_1}$  and  $b_{l_1}$  are initialized to zero matrix 0. The training algorithm of LSA deep neural network is shown in Algorithm 1, which includes two stages: pre training and fine-tuning. In this algorithm, the traffic data generated by grid  $(m, n)$  and its neighbors at time  $t$  forms a data slice patch  $(d_{m,n,t})$ .  $\tilde{X}_i$  refers to  $X_i$  with errors. The 1 ~ 4 line of Algorithm 1 generates input data for the first layer of LSA. The 5th and 6th lines of the algorithm are initialization operations. The seventh line uses data with errors (that is  $\tilde{X}_i$ ) as input for pre training the first layer network. The reconstruction loss is defined as the cross entropy in, as follows:

$$L(X, Z) = \sum x \log(x) + (1 - x) \log(1 - z) \quad (5)$$

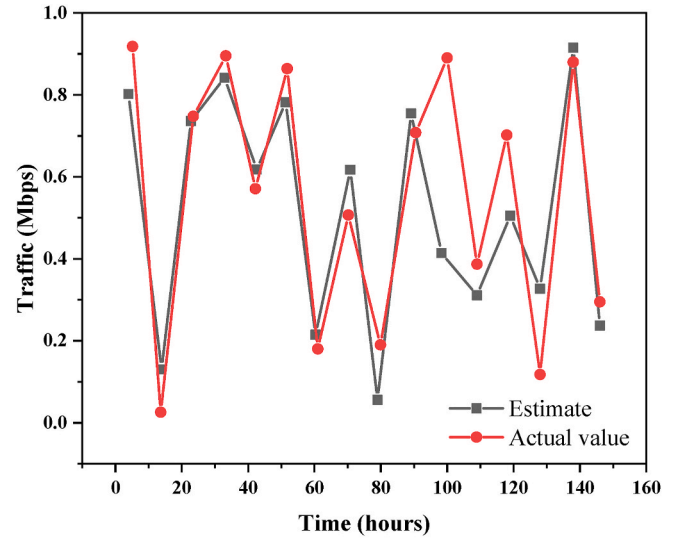
The minimization of reconstruction loss was achieved using a stochastic gradient descent algorithm. The eighth line of the algorithm uses  $X_{L_1}$  to generate the input  $X_{L_2}$  for the second layer. The 9 ~ 12 line is the pre training process from layer 2 to layer  $j$ . After pre training the  $j$ th layer, the input  $X_{l_{j+1}}$  (that is  $X_{l_{j+1}} = enf_{l_j}(X_{l_j})$ ) of the  $j+1$  layer can be obtained. After pre training all layers of the network, unfold GSA and LSA for fine-tuning. As mentioned above, Recurrent Neural Networks (RNNs) are used for time modeling and prediction, taking encoded representations obtained from mixed spatial models as inputs. However, standard RNNs are not suitable for modeling long-term dependencies. LSMN can capture long-term temporal dependencies, allowing models to learn how to forget hidden states and how to update the current state. The structure of LSMN is shown in Fig. 3.

LSMN consists of a memory unit  $c_t$ , an input module  $g_t$ , an output module  $h_t$ , and three logic gates (input gate  $i_t$ , output gate  $o_t$ , and forget gate  $f_t$ ). Among them,

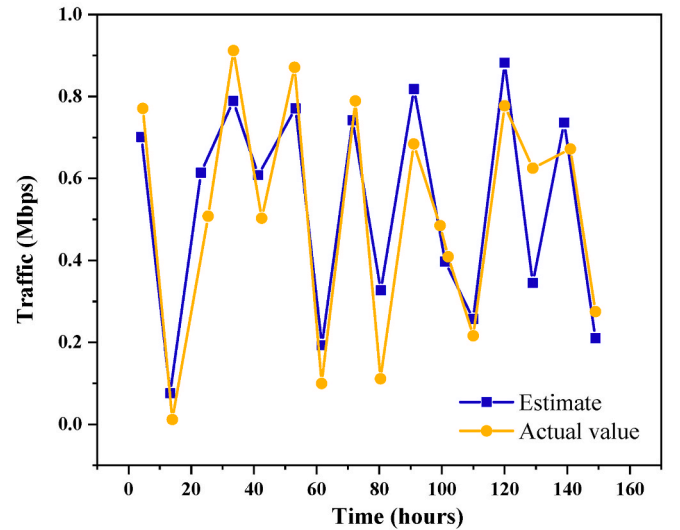
$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i) \quad (6)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f) \quad (7)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o) \quad (8)$$



(a) Uplink traffic



(b) Downlink traffic

Fig. 4. Comparison between predicted and actual traffic values.



**Table 1**

Comparison of results of mean square error, mean absolute error, and cross entropy.

|                        | A certain grid                |        |                             |        | Average of all grids               |        |                             |        |
|------------------------|-------------------------------|--------|-----------------------------|--------|------------------------------------|--------|-----------------------------|--------|
|                        | Based on deep learning models |        | Based on time series models |        | Based on deep learning, this model |        | Based on time series models |        |
|                        | up                            | down   | up                          | down   | up                                 | down   | up                          | down   |
| Mean squared error     | 0.0224                        | 0.0340 | 0.0622                      | 0.0702 | 0.0297                             | 0.0356 | 0.0533                      | 0.0588 |
| Average absolute error | 0.1128                        | 0.1303 | 0.1828                      | 0.1971 | 0.1197                             | 0.1401 | 0.1722                      | 0.1830 |
| Cross entropy          | 0.4111                        | 0.3627 | 0.6538                      | 0.6087 | 0.4516                             | 0.5101 | 0.6022                      | 0.5795 |

$$g_t = \sigma(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (9)$$

$$c_t = f_t \odot c_{t-1} \oplus i_t \odot g_t \quad (10)$$

$$h_t = o_t \odot \phi(c_t) \quad (11)$$

$\phi(x)$  is a hyperbolic tangent function. The symbols  $\odot$  and  $\oplus$  represent point multiplication and vector summation, respectively. The memory unit updates the hidden state by combining the previous grid state, current input, and previous output. The forgetting gate determines whether information should be forgotten or remembered. How the output gate learning memory unit affects the hidden state. In order to predict the future value  $d_{m,n,t'+1}$  of grid  $(m,n)$ , data slices from the previous  $T$  time intervals will be used as inputs and then encoded by GSA and LSA. For each moment, the values of  $t \leq t'$ ,  $d_{m,n,t}$  and  $patch(d_{m,n,t})$  will be concatenated into a vector. In this way, we obtained the time series of the vector, and the LSMN module processed the sequence and predicted  $d_{m,n,t'+1}$ .

#### 4. Results and discussion

Both deep learning based and time series analysis based models are implemented in the Keras function library to evaluate the average squared error, average absolute error, and cross entropy of the two algorithms. Select  $11 \times 11$  grids centered on the target grid as neighboring grids. Fifteen grids were randomly selected for testing, and for each network  $(m,n)$ , the data was divided into a training set and a testing set. Firstly, present the experimental results of the predictive performance based on deep learning models. As shown in Fig. 4, Fig. 4 (a) shows the predicted and actual values for the uplink, and Fig. 4 (a) shows the results for the downlink. As shown in Fig. 4, the trend between the predicted values and the actual values is relatively consistent. Table 1 shows the comparison of average squared error, average absolute error, and cross entropy for downlink/uplink based on deep learning models and time series models. According to Table 1, the deep learning based model outperforms the time series based model in all metrics. The experimental results demonstrate the effectiveness of applying deep learning models to wireless network data analysis.

According to data analysis, it can be seen that the prediction of the system is effectively improved by 99 %.

#### 5. Conclusion

The author proposes research on spatiotemporal data modeling and prediction algorithms in intelligent management systems, including a new self coding based spatial modeling model and a time model based on long short-term memory networks. Firstly, the wireless network data (mainly the data of cellular mobile networks) obtained by Internet service providers are preliminarily analyzed, revealing that the data of adjacent base stations have temporal and spatial correlations. Then, a hybrid deep learning model for spatio-temporal prediction is established, and a new spatial model training algorithm is proposed. Finally, experiments were conducted using wireless network datasets to evaluate the performance of the model. The experimental results show that compared with existing methods, the author's deep model significantly

improves prediction accuracy and has broad application prospects.

#### Declaration of competing interest

The authors declare that they have no competing interests.

#### Data availability

No data was used for the research described in the article.

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